

Wine Quality Exploratory Data Analysis

~Exploratory Data Analysis (EDA) Summary Report

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Objective

The objective of this project was to explore the Wine Quality dataset using Exploratory Data Analysis (EDA). The project focused on understanding the dataset, performing descriptive statistical analysis, identifying relationships between different features, detecting outliers, filtering and sorting the data, and visualizing important patterns that influence wine quality.

Dataset Information

Dataset Name: Wine Quality Dataset

Source: Kaggle

Dataset Link:

<https://www.kaggle.com/datasets/yasserh/wine-quality-dataset>

Dataset Description

The Wine Quality dataset contains various physicochemical properties of wine samples, such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, alcohol content, sulphates, pH, density, and quality score. The dataset is commonly used for data analysis to understand the factors that affect the overall quality of wine.

Data Exploration

The dataset was explored using functions such as `head()`, `shape`, `info()`, and `describe()`. These functions helped understand the structure of the dataset, identify data types, examine the number of records, and generate descriptive statistics before performing further analysis.

Data Cleaning

The dataset was checked for missing values and duplicate records. Missing values were handled wherever required, and duplicate entries were removed to improve the quality of the dataset. These preprocessing steps ensured that the data was clean and ready for analysis.

Statistical Analysis

Descriptive statistical measures such as mean, median, mode, variance, and standard deviation were calculated for the numerical columns. These measures helped understand the distribution, spread, and variation of different wine characteristics.

Correlation Analysis

The correlation matrix was calculated to understand the relationship between different numerical features. A correlation heatmap was generated to identify highly correlated variables. This analysis helped determine which features have a stronger influence on wine quality.

Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. Box plots were created to identify extreme values in numerical columns. Detecting outliers helped in understanding unusual observations that could influence the analysis.

Filtering and Sorting

Filtering and sorting operations were performed to analyze specific groups of wine samples based on different feature values. These operations made it easier to compare records and understand the characteristics associated with different quality levels.

Data Visualization

Different visualizations were created to understand the dataset more effectively. These included Bar Chart, Line Chart, Histogram, Scatter Plot, Correlation Heatmap, and Box Plot. The visualizations helped identify data distributions, feature relationships, and quality trends among wine samples.

Key Insights

The dataset was successfully explored and analyzed using EDA techniques. Statistical analysis provided valuable information about the numerical features, while correlation analysis identified important relationships between variables. Outliers were detected using the IQR method, and visualizations clearly illustrated the characteristics that influence wine quality.

Conclusion

This project provided practical experience in performing Exploratory Data Analysis using the Wine Quality dataset. It covered data exploration, descriptive statistics, correlation analysis, outlier detection, filtering, sorting, and visualization techniques. The project demonstrated how EDA can be used to identify important factors affecting wine quality and support data-driven analysis.

Tools Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
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Project Outcome

Through this project, I gained hands-on experience in:

- Data Exploration
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Descriptive Statistics
- Correlation Analysis

- Outlier Detection
 - Data Filtering and Sorting
 - Data Visualization
 - Interpreting Analytical Results
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